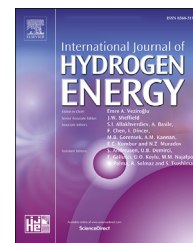


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CaH₂-assisted structural engineering of porous defective graphitic carbon nitride (g-C₃N₄) for enhanced photocatalytic hydrogen evolution

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HIGHLIGHTS

- A strategy with the assist of CaH₂ is proposed to synthesize defective g-C₃N₄.
- Various types of defects are introduced into catalyst (MM-H) in one step.
- The MM-H shows manipulated porosity and modulated band structure.
- The MM-H exhibits an over 7 times enhancement in photocatalytic H₂ generation.

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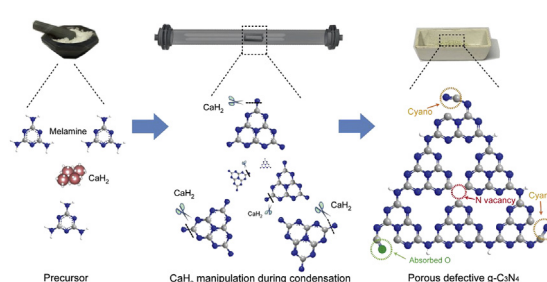
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GRAPHICAL ABSTRACT



ABSTRACT

The practical applications of graphitic carbon nitride (g-C₃N₄) for photocatalytic hydrogen evolution is strictly hindered by the low surface area, poor light harvesting capability and detrimental recombination of photoexcited charge carriers. Herein, using melamine as precursor and metal hydride (i.e., CaH₂) as active agent, we facilely incorporate various types of defects (i.e., nitrogen (N) vacancies (V_N), cyano groups (C≡N) and surface absorbed oxygen species (O_{abs})) into g-C₃N₄ within a single step. The as-prepared material (denoted as MM-H) exhibits narrowed bandgap, promoted photoexcited electron-hole separation rate and facilitated charge transfer kinetics with enlarged BET surface area and massive

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porosity. As a result, a prominently enhanced photocatalytic H₂ productivity efficiency (1305.9 $\mu\text{mol h}^{-1}\text{g}^{-1}$) is shown on MM-H. This performance is better than that of g-C₃N₄ with CaH₂ post-treatment (617.3 $\mu\text{mol h}^{-1}\text{g}^{-1}$) and raw bulk-C₃N₄ (178.2 $\mu\text{mol h}^{-1}\text{g}^{-1}$). This work opens up a new dimension for designing high performance g-C₃N₄-based catalysts targeting various photocatalytic processes.

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Introduction

The environmental issues stemming from consuming of fossil fuels have been paid interdisciplinary attentions in recent years. As a clean and sustainable energy carrier, hydrogen (H₂) is a potential alternative energy source with zero emission [1–3]. Compared with thermocatalysis and electrocatalysis H₂ production processes, photocatalytic H₂ evolution is an attractive strategy to produce storable chemical fuel only at the consumption of inexhaustible solar energy, in which searching for high efficient and low-cost photocatalysts is the crucial step and has been widely studied for decades [3–8].

Graphitic carbon nitride (g-C₃N₄) is a promising candidate in both photosynthesis and photocatalytic fields owing to the natural abundance of its precursor, biocompatibility, suitable energy level for water reduction/oxidation, chemical/thermal stability, and non-toxicity [9–13]. Generally, g-C₃N₄ can be prepared by thermal polymerization of inexpensive nitrogen-riched precursors (such as melamine, dicyanamide, urea or thiourea). However, it is usually an incomplete process [14–16]. As a result, the unreacted amino groups (NH_x groups) in the g-C₃N₄ framework are potential recombination center for photoexcited charge carriers, which severely hinder their separation (Typically, charge carriers recombination happens within 10^{−12}–10^{−9} s), and weaken the photocatalytic performance [17–19]. So far, much efforts have been undertaken to circumvent these challenges for g-C₃N₄, such as nanostructure engineering [4], heteroatoms doping [20–24], exfoliation of bulk material to 2-dimensional nanosheets [25–27], and heterostructure construction by co-catalyst [28–34].

Beyond that, the introduction of catalytically relevant defects on the surface of g-C₃N₄ via post-treatment is recently recognized as a prevailing way to enhance the separation efficiency of charge carriers and intrinsic photocatalytic performance [11,35]. The defects in the tri-s-triazine repeating units are desirable to manipulate the band level, and also offer extra reactive sites for photocatalytic reaction. For examples, the formation of nitrogen vacancies (V_N) in g-C₃N₄ was shown to substantially enhance the photocatalytic performance [36]. In addition, it has been reported that the additive of potassium isothiocyanate could replace the surface amino groups (–NH₂ and –NH) of g-C₃N₄, which brought in abundant cyanamide defects (C≡N) as coordination groups and catalytically active sites [37]. Moreover, g-C₃N₄ modulated by NaBH₄ could also produce p-type electron-accepting defects of C≡N in n-type g-C₃N₄ bulk [38], leading to the formation of p-n homojunctions and the suppression of charge carriers recombination. Alkali-

assisted thermal polycondensation can also bring C≡N defects into g-C₃N₄, which could effectively adjust the crystallinity and band structure of the materials [39].

Nevertheless, to date, the approaches applied for defects engineering in g-C₃N₄ are usually complex and time-consuming, because most of them require multiple steps post-treatment. Furthermore, they inherently suffer from very limited control over species and abundance of defects for selectively incorporation. Therefore, it is highly desirable to develop economical and effective methods to introduce multi-type defects with optimal concentration into g-C₃N₄, which can be expected to have promoted performance and potential for massive production.

In this work, we reported a feasible route to prepare porous defective g-C₃N₄ materials, wherein different kinds of defects (i.e., V_N, C≡N group and surface absorbed oxygen species (O_{abs})) with optimal concentration and uniform distribution were implanted into g-C₃N₄ framework during a one-pot thermal polymerization and deamination reactions between melamine and CaH₂ (Fig. 1). Excitingly, the as-prepared photocatalyst (denoted as MM-H, where MM stands for melamine, and H represents CaH₂) demonstrates increased specific surface area with formation of massive porosity, modulated band structure, facilitated photoexcited hole-electron pairs separation, and accelerated charge carrier transportation, which strengthens the versatility of this method. Compared to bulk g-C₃N₄ (CN) and g-C₃N₄ with post-treatment of CaH₂ (CN-H), the MM-H shows an apparent enhancement of photocatalytic H₂ evolution performance using methanol as a sacrificial agent. The methodology adopted here may shed light on a new direction for design of g-C₃N₄.

Results and discussion

Thermogravimetric-differential scanning calorimetry analysis (TG-DSC) was used to determine the annealing parameters and analyze the reaction during thermal polycondensation process to form CN, CN-H and MM-H, respectively. The experiment was performed from 25 °C to 600 °C with a N₂ flow rate of 10 mL min^{−1}. As shown in Fig. 2a and Fig. S1, the condensation of the melamine to form g-C₃N₄ network starts at 260 °C with only one significant exothermic peak at ~300 °C. Similarly, there is only one exothermic peak located at ~400 °C for CN-H sample, which can be explained by chemical reaction between CaH₂ and g-C₃N₄. Differently, the direct reaction between melamine and CaH₂ to form MM-H demonstrates a distinct two-step process. Before 350 °C, the

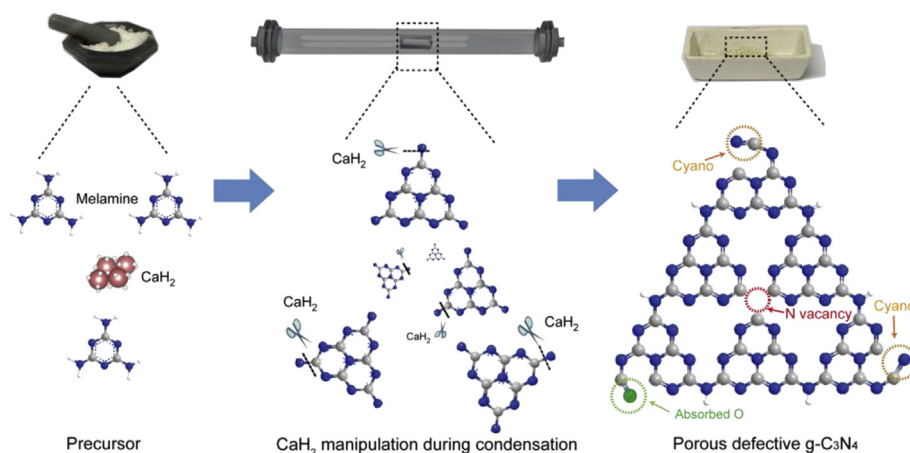


Fig. 1 – The proposed one-pot synthetic route of porous defective g-C₃N₄ (MM-H).

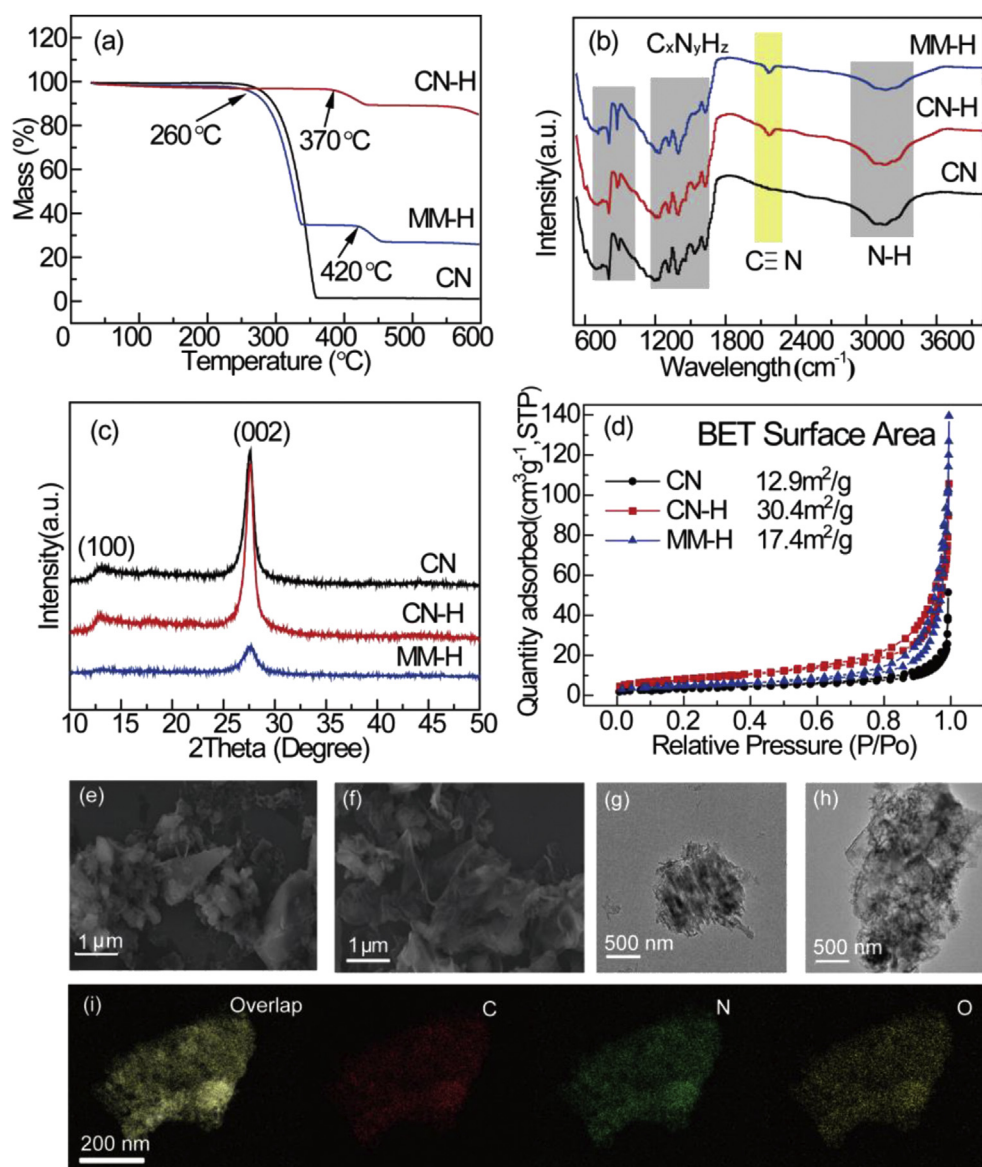


Fig. 2 – (a) TGA curves in pure nitrogen atmosphere; (b) FTIR spectra; (c) XRD patterns; (d) N₂ adsorption-desorption isotherms of CN, CN-H and MM-H samples. (e-f) SEM images of (e) CN and (f) MM-H. (g-h) TEM images of (g) CN and (h) MM-H. (i) The corresponding C, N and O EDX mapping of MM-H.

TGA curves for MM-H and CN are almost overlapped. However, considering the weight ratio of melamine vs. CaH_2 of 3:1, it is reasonable to propose that the g- C_3N_4 starts to react with CaH_2 (at the weight loss level of ~35%) before the completion of polycondensation. As seen in DSC curve in Fig. S1, there are two exothermic peaks appearing on MM-H sample, which are caused by the polymerization of the melamine (~300 °C, identical to CN) and the reaction of *in-situ* formed g- C_3N_4 with CaH_2 (~450 °C), respectively.

The vibration of various chemical bonds in CN, CN-H and MM-H were investigated by Fourier transform infrared spectroscopy (FTIR) spectroscopy. As shown in Fig. 2b, the vibration peak at 810 cm^{-1} is owing to the representative out-of-plane bending of tri-s-thiazine subunits, indicating the ordered in-plane structure [37]. The presence of peaks in the region from 1000 to 1700 cm^{-1} can be ascribed to the stretching vibration of C–N heterocycle and aromatic heterocyclic systems, revealing a relatively stable polymeric melon framework for all the samples [40–42]. Specifically, the peaks at 1320 cm^{-1} and 1550 cm^{-1} are attributed to the stretching vibration of C–N (sp^3) and C=N (sp^2), respectively [27,30,43]. The peak at 1235 cm^{-1} represents the stretching vibration of C–NH–C bond, which is a basic subunit of g- C_3N_4 . On MM-H, this peak slightly shifts to higher frequency region (1255 cm^{-1}), indicating the possible formation of C–O with strong electron-attracting effect [44]. Moreover, a distinct peak centered at 2180 cm^{-1} corresponds to an asymmetric stretching vibration of C≡N which is derived from the deprotonating of amino groups [45]. This peak can be observed on both MM-H and CN–H, but is absent on CN, suggesting that the additive of CaH_2 is the reason for the formation of C≡N. The cyano groups are believed to form during the opening of tri-s-triazine heterocycles of g- C_3N_4 along with the lattice V_N formation [40]. It was reported that the g- C_3N_4 is an n-type semiconductor and the terminal –C≡N triple bond could act as strong electron acceptors (p-type) to form the intrinsic p-n homojunctions [38,46,47]. The broad peaks between 3000 and 3500 cm^{-1} is due to the stretching vibration of N–H, suggesting the existence of amino group (N–H) stemming from the incomplete polymerization [40,41]. Obviously, MM-H shows a much weaker peak intensity of N–H, indicative of its more complete polycondensation. All in all, the results of FTIR imply that the CaH_2 molecule could react with the amine group of melamine-derived intermediates to generate abundant C≡N and V_N , and facilitate the deamination and polymerization process.

The phase evolution of all photocatalysts was characterized by powder X-ray diffraction (XRD) analysis and shown in Fig. 2c. CN exhibits two apparent diffraction peaks located at $2\theta = 13.1^\circ$ and 27.4° , respectively. The broader peak at 13.1° originates from the (100) plane of g- C_3N_4 and can be ascribed to the inter-planar structural packing of heptazine units [48,49]. The relatively sharper diffraction peak at 27.4° should be indexed to the (002) plane of g- C_3N_4 and represents the interlayer stacking of π - π conjugated aromatic units [46]. The XRD patterns of CN–H, MM-H materials are very similar to those of CN, confirming that the architecture of tri-s-triazine framework is maintained after reaction with CaH_2 . However, unlike the CN and CN–H materials, both peaks belonging to (100) and (002) are significantly broadened and

weakened on MM-H, indicating that the ordered and periodic framework of polymeric melon were broken during the thermal polymerization process in presence of CaH_2 . It is partially due to the formation of high concentration of in-plane V_N and the exfoliation of bulk g- C_3N_4 [25,50], which may further influence the band levels. More interestingly, the lateral (002) peak of MM-H shifts to a higher 2θ angle at $\sim 27.7^\circ$, suggesting that the MM-H has a smaller interlayer stacking distance between g- C_3N_4 sheets after the one-pot treatment of CaH_2 , which is probably due to the facilitated condensation effect of CaH_2 [51].

The N_2 adsorption-desorption isotherms were carried to confirm the improved specific surface area after *in-situ* CaH_2 treatment (Fig. 2d). The corresponding pore size distribution curves are shown in Fig. S2. The N_2 adsorption-desorption curve for MM-H sample displays a type-IV isotherm accompanied with a type-H3 hysteresis loop, suggesting the possible presence of mesoporous structure originating from the stacking of CN. The specific surface areas of CN, CN–H and MM-H are $12.9\text{ m}^2/\text{g}$, $17.4\text{ m}^2/\text{g}$ and $30.4\text{ m}^2/\text{g}$ with the corresponding pore volume of $0.079\text{ cm}^3/\text{g}$, $0.136\text{ cm}^3/\text{g}$ and $0.185\text{ cm}^3/\text{g}$, respectively. Evidently, the MM-H shows a promoted BET surface area and enhanced porosity. The morphology of as-synthesized MM-H material was further characterized using SEM and TEM (Fig. 2e–i). The CN material prepared from direct polymerization of melamine (without CaH_2) is readily aggregated, as shown in Fig. 2e (SEM) and Fig. 2g (TEM). Such material generally possesses low specific surface area and is striking lack of active sites, undoubtedly restricting its photocatalytic activity. In comparison, MM-H exhibits a porous structure with ubiquitous stacking cloud-like morphology (Fig. 2f and h and S3). Based on the element mapping analysis results (Fig. 2i), the elements of C, N and O were uniformly distributed in MM-H.

Besides to microstructure, the surface defects should play vital roles in the photocatalytic process. To understand the role of defects on band structure engineering of the materials at atomic scale, X-ray photoelectron spectroscopy spectra (XPS) were further employed. The full spectra of each sample (Fig. S4) reveal that the CN, CN–H and MM-H photocatalysts mainly consist of C and N elements. The existence of trace amount of oxygen species could be owing to the surface O heteroatoms or other adsorbed oxygen-containing species (e.g., H_2O). The high-resolution C 1s, N 1s and O 1s spectra of each material were shown in Fig. 3a–c, respectively. The high resolution C 1s spectrum is deconvoluted into four peaks at 290.2 , 288.2 , 286.5 , and 284.9 eV which can be ascribed to the C–O bond, sp^2 -bonded N–C=N coordination, C≡N/C–NH_x and C–C/C=C, respectively [52]. Obviously, the peak intensity at 286.5 eV is raised for MM-H, as compared to those of CN–H and CN. Combined with the results of FTIR, the evolution of XPS survey could be regarded as an evidence supporting the formation of C≡N bond, because C≡N and C–NH_x share the similar C 1s binding energy [40]. The ascending intensity of peak at 290.2 eV for MM-H suggests the potential incorporation of O heteroatoms, which is consistent with the high-resolution O 1s XPS spectra (Fig. 3b). The binding energy peak of oxygen shifts from 532.2 eV for CN to 531.85 eV for MM-H, implying the generation of adventitious carboxyl oxygen-containing group after CaH_2 treatment [42,53]. This

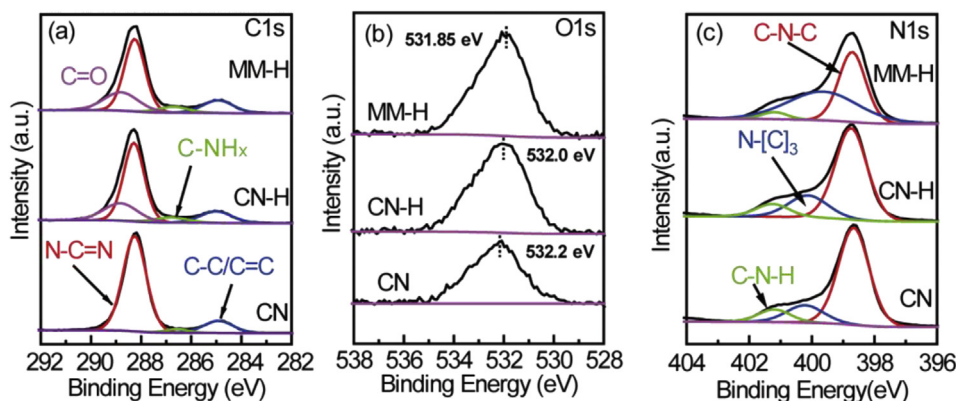


Fig. 3 – High resolution (a) C 1s, (b) O 1s and (c) N 1s XPS spectra of CN, CN-H and MM-H photocatalysts.

conclusion can also be confirmed by the result of SEM-EDS (Table S1) in which the remarkably enhanced value of O content (11.89%) is observed on MM-H. The high resolution N 1s spectrum shown in Fig. 3c could be resolved into three components located at ~401.2 eV, 399.9 eV and 398.7 eV, which responds to the C–N–H, tri-coordinated (N-[C]₃) and bi-coordinated (C–N=C) in the heptazine framework, respectively [42]. Compared to the spectra of CN, the XPS peak of tri-coordinated N in MM-H shifts toward to lower binding energy region (from ~400.2 eV to 399.6 eV), owing to the missing of N atoms [54]. And the decreased intensity of C–N–H on MM-H buttresses the promoted polycondensation effect. The elemental composition analysis of all photocatalysts determined by XPS is shown in Tables S2–3. The ratio of N-[C]₃/C–N=C shows a rising tendency with order of 0.25 (CN) < 0.29 (CN-H) < 1.08 (MM-H). The value of C/N increases from 0.80 for CN to 0.83 for MM-H, with the content of N decreasing from 51.54% (CN) to 50.71% (MM-H). Also, the ratio of C–N=C/C declines from 0.154 (CN) to 0.099 (MM-H). All these results collectively confirm the formation of C≡N groups and V_N. In addition, the results of organic element analysis (OEA) is consistent with XPS analysis (Fig. S5), and the improved value of C/N from 0.561 to 0.581 was observed on MM-H.

To evaluate the band structure engineering effect of CaH₂, the electron paramagnetic resonance (EPR) spectra was performed at room temperature. As shown in Fig. 4a, the CN exhibits a paramagnetic absorption signal with the g value of 2.0034 which is owing to the unpaired electrons on C atoms in π -conjugated aromatic rings [48]. Obviously, the MM-H catalyst shows a significantly strengthened EPR signal intensity with the g value shift to 2.0023. This variation should be explained by the synergistic effect of redistribution of π -electrons due to the introduction of surface absorbed O species and cyano-terminal groups, collectively leading to the extension of electronic delocalization and benefited charger transfer [37].

The optical bandgap of the photocatalysts were studied by UV–vis diffuse reflectance spectra (UV–vis DRS) (Fig. 4b) and the corresponding Kubelka-Munk (KM) plot is shown in inset of Fig. 4b. It is clear that all samples show typical semiconductor properties of polymeric carbon nitride [9,55]. The main absorption peak at $\lambda < 440$ nm and the absorption tail starting at $\lambda = 520$ nm originate from the $\pi \rightarrow \pi^*$ electron

transitions and $n \rightarrow \pi^*$ electron transitions, respectively [56]. (Fig. 3e). The Kubelka-Munk method was further employed to derive the optical bandgap according to the equation: $\alpha h\nu = A(h\nu - E_g)^{n/2}$, where α , A, h, and ν are the absorption coefficient, the constant, the Planck constant and the frequency of the light, respectively. n is equal to 1 for direct bandgap semiconductor such as g-C₃N₄ [57]. Compared with CN with bandgap of 2.78 eV, the MM-H exhibits a decreased bandgap value of 2.72 eV, indicating its better light harvesting capability. The absolute position of the valence band (VB) of each sample was derived from valence band (VB) XPS spectra (Fig. 4c). The CN, CN-H and MM-H demonstrate an estimated valence band position of ~1.72, 1.68, and 1.59 eV, respectively. Therefore, the conduction band (CB) position of CN, CN-H and MM-H samples were calculated to be -1.04, -1.1 and -1.13 eV (Fig. 4d). On one hand, from the perspective of thermodynamic, the negatively shifted CB position of MM-H is expected to provide more driving force for proton reduction to H₂ [56]. On the other hand, the VB position of 1.59 eV is highly promising for photocatalytic oxygen evolution reaction (OER) @ 1.23 V vs. NHE) and further the overall water splitting. Because the photoexcited hole leaving in VB is not only favorable for OER based on theoretical calculation, but also can forbid the side reaction of H₂O₂ production (1.8 V vs. NHE).

As aforementioned theory, the capability for light-absorption, the separation of photo-generated charge carriers and transfer efficiency are decisive factors for photocatalytic performance [58,59]. The introduction of C≡N in g-C₃N₄ is expected to form intrinsic p-n homojunctions which will suppress the recombination of photoexcited charge carriers. As characterized by photoluminescence (PL) spectroscopy (Fig. 5a), CN appears a broad PL peak centered at about 440 nm under 420 nm excitation at room temperature, which is attributed to the recombination of the photogenerated carriers with emission photon energy equal to its bandgap energy [42]. The MM-H catalyst presents a much quenched steady-state PL intensity, proving its improved separation efficiency of photoexcited charge carriers. Also, the PL peak maximum of MM-H significantly redshifts to longer wavelength (~450 nm), consistent with the result that the MM-H possesses a narrowed bandgap benefiting for light harvesting. The measurement of time-resolved photoluminescence (TRPL) charge carrier lifetime was further performed for all

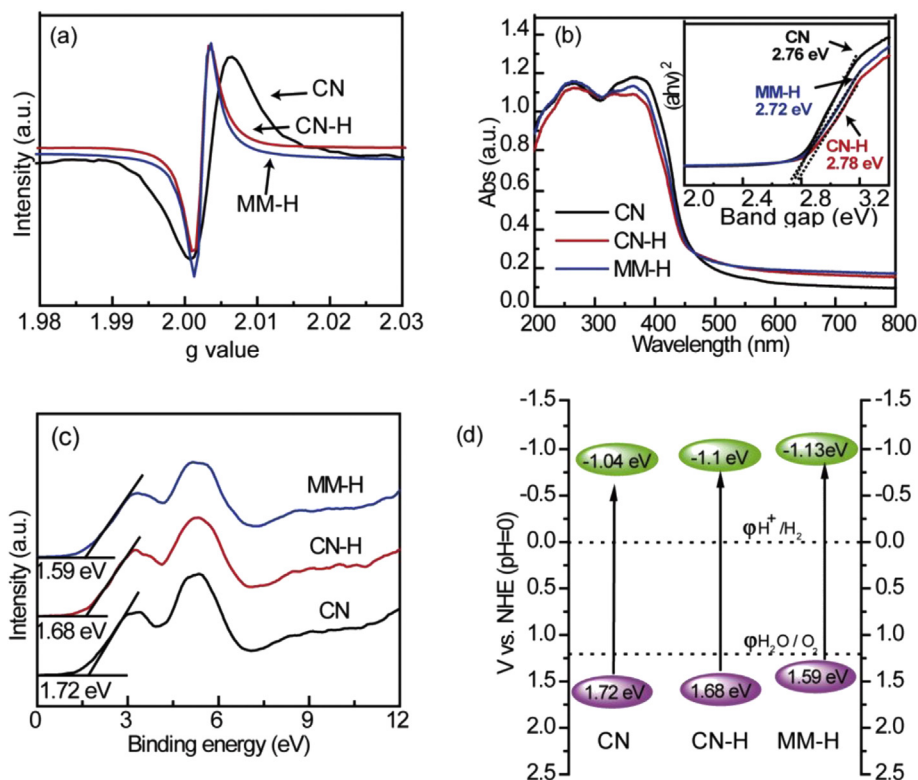


Fig. 4 – (a) Electron paramagnetic resonance (EPR) signals; (b) UV-VIS diffuse reflectance spectroscopy (DSR) spectra (inset is the corresponding Kubelka-Munk transformed reflectance spectra); (c) Valence Band (VB) XPS spectra and (d) Estimated electronic band level of CN, CN-H and MM-H photocatalysts.

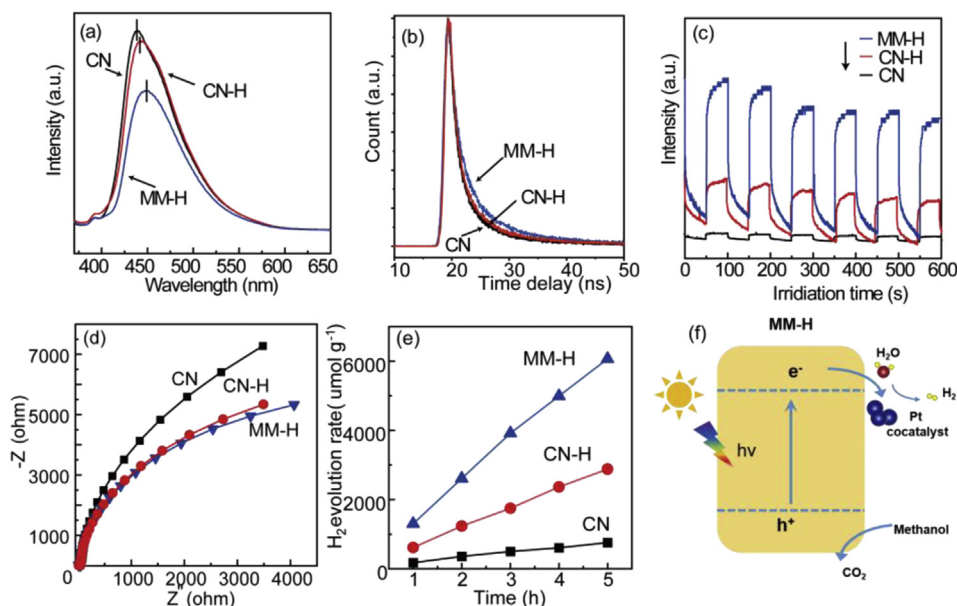


Fig. 5 – (a) Photoluminescence emission spectra (PL excitation: 420 nm) (b) Time-resolved photoluminescence (TRPL) charge carrier lifetime plot (The instrument response function (IRF) is ~0.02 ns) (c) Transient photocurrent density recorded under visible irradiation (potential: -0.4V vs. Ag/AgCl) (d) Nyquist plots of CN, CN-H, MM-H photocatalysts, respectively; (e) The plots of time dependent hydrogen production rate of 3%wt Pt/CN, CN-H and MM-H photocatalysts, respectively. (f) Illustration of photocatalytic H₂ production mechanism.

samples and the plot is shown in Fig. 5b. Compared to CN (12.7 ns), the MM-H demonstrates a prolonged PL lifetimes (31.3 ns), suggesting that the additive of CaH_2 can elongate the lifetime of charge carriers and enhance their efficiency for the following water-splitting reactions.

To better quantify the charge separation efficiency of MM-H, the transient photocurrent responses of the as-prepared photocatalysts were further measured. MM-H (Fig. 5c) shows the highest photocurrent intensity, which further buttresses the conclusion that MM-H has the best charge separation and transfer among the three samples. Additionally, the electrochemical impedance spectroscopy (EIS) was carried to further confirm their different charge transfer properties [60]. The smallest arc radius for MM-H photocatalyst in Nyquist plot certifies its smallest charge transfer resistance and fastest interfacial charge transfer, as compared to CN and CN-H (Fig. 5d). The as-mentioned data collectively prove that the capability of light-absorption and efficiency of charge carrier separation of MM-H are promoted owing to incorporation of defects. Fig. 5e shows the photocatalytic H_2 -production activities of CN, CN-H and MM-H photocatalysts. The MM-H photocatalyst exhibits a best activity of $1305.9 \mu\text{mol h}^{-1}\text{g}^{-1}$, which is seven times and two times of efficiency from CN ($178.2 \mu\text{mol h}^{-1}\text{g}^{-1}$) and CN-H ($617.3 \mu\text{mol h}^{-1}\text{g}^{-1}$), respectively. This enhancement is comparable to other representative works shown in literatures (Table S4). A speculated mechanism for photocatalytic hydrogen evolution reaction on MM-H composite is illustrated in Fig. 5f. Under light irradiation, the photocatalyst could harvest light and yield the photoinduced electrons and holes in the conduction and valence band, respectively. Then, the electrons on the MM-H surface migrate to the surface deposited cocatalyst, couple with the Pt particles, and further react with hydrogen ions to produce H_2 . Meanwhile, the holes react with methanol through oxidation to suppress the recombination of photo-excited hole-electron pair. Herein, it is reasonable to make the conclusion that the porous MM-H with homogenous distribution of three types of defect is a promising photocatalyst for photocatalytic hydrogen evolution.

Conclusions

To summarize, we developed a facile process to fabricate defective porous g- C_3N_4 photocatalyst by taking advantage of one-pot thermal treatment of melamine with CaH_2 under nitrogen atmosphere. An achievement of “one stone, three birds” was obtained on MM-H material from several perspectives. First of all, according to FTIR and XPS results, the formation of n-type $\text{C}\equiv\text{N}$ was confirmed in p-type g- C_3N_4 framework. The generation of p-n homojunctions is highly desirable for light harvesting, photogenerated electron-hole separation and charge transfer. Secondly, on the basis of the XRD, XPS, OEA and EPR analyses, the addition of CaH_2 can simultaneously partially remove N-atoms to form V_N and insert surface absorbed O species. Both of them benefit the charge transfer. Thirdly, the addition of CaH_2 facilitates the precursor polymerization and exfoliation of stacking of g- C_3N_4 , resulting in larger BET surface area and mesoporous structure. Benefiting from these merits, the photocatalytic H_2

productivity of MM-H sample ($1305.9 \mu\text{mol h}^{-1}\text{g}^{-1}$) is shown to be seven times of activity from raw CN ($178.2 \mu\text{mol h}^{-1}\text{g}^{-1}$), and over two times of activity from CN with CaH_2 post-treatment ($617.3 \mu\text{mol h}^{-1}\text{g}^{-1}$). This work demonstrates a novel tactic to prepare high performance g- C_3N_4 based photocatalyst for efficient hydrogen generation.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2020.05.044>.

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